

A
Project Report on
Predicting Autism Spectrum Disorder Using Machine Learning
Submitted in partial fulfilment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

By

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Technology**

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CERTIFICATE

This is to certify that the project report entitled “**PREDICTING AUTISM SPECTRUM DISORDER USING MACHINE LEARNING**” is a bonafide work done by **Likhith Bathoju (22R11A6750), Burkala Nithin(22R11A6753), Tuniki Santhoshini (22R11A6783)**, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in “**Computer Science and Engineering (Data Science)**” from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026.

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We, **Likhith Bathoju, Burkala Nithin, Tuniki Santhoshini** bearing Roll Nos. **22R11A6750, 22R11A6753, 22R11A6783** hereby declare that the project report entitled **“Predicting Autism Spectrum Disorder using Machine Learning”** is done under the guidance of **Mrs.Jayasree, Assistant Professor**, Department of Computer Science and Engineering (Data Science), Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in CSE(Data Science)**.

This is a record of bonafide work carried out by us in **Geethanjali College of Engineering and Technology** and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other University or Institute for the award of any other degree or diploma.

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Correlation Between The Major Project and the Program Outcomes (Po's, Program Specific Outcomes (Pso's))

IV YEAR II SEM SECTION – B BATCH NO:15		TITLE OF THE PROJECT
ROLL NO	NAME	Predicting Autism Spectrum Disorder using Machine Learning
22R11A6750	Likhith Bathoju	
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Table 1: Major project coorelation with appropriate PO's and PSO's
(Please specify level of correlation, High – 3, Medium – 2 and Low – 1)

PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
3	3	3	3	3	2	2	3	2	2	2	3	3	2

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Mrs.Jayasree



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Sustainable Development Goals (SDG) Table for < PREDICTING AUTISM SPECTRUM DISORDER USING MACHINE LEARNING >

SDG No.	SDG NAME	Relevance to your project	How your System contributes
SDG 9	Industry, Innovation and Infrastructure	The project uses advanced machine learning models for solving healthcare problems	Promotes innovation in healthcare technology by providing an intelligent ASD prediction system
SDG 17	Partnerships for the Goals	Encourages collaboration between technology and healthcare domains	Supports integration of AI tools into healthcare systems for better diagnosis and research

ABSTRACT

The rapid advancement of artificial intelligence and machine learning technologies has enabled the development of intelligent systems capable of analyzing complex healthcare data for early disease detection. Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects communication, social interaction, and behavior, making early diagnosis essential for timely intervention and improved developmental outcomes. Therefore, developing an efficient and automated system for early ASD detection has become an important research problem in the field of healthcare analytics and machine learning.

This project proposes a machine learning–based system to predict Autism Spectrum Disorder by analyzing behavioral screening responses and demographic data. Advanced machine learning models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN) are used for classification and prediction. The trained models generate probability scores, which are further enhanced using a hybrid rule-based decision mechanism based on A1–A10 behavioral screening scores. The performance of the models is evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliable prediction outcomes.

The proposed approach aims to provide an efficient and accessible solution for early ASD screening and decision support. The system offers an automated and scalable framework for predicting ASD risk, with potential applications in healthcare support systems, clinical decision-making, educational planning, and early intervention strategies. This work contributes to improving early detection, enhancing accessibility to screening tools, and supporting informed decision-making for healthcare professionals, educators, and caregivers.

Keywords: Autism Spectrum Disorder (ASD), Machine Learning, XGBoost, Random Forest, Artificial Neural Network (ANN), Healthcare Analytics, Behavioral Screening, Data Preprocessing, Feature Selection, Label Encoding, Standardization, Hybrid Prediction System, Accuracy, Precision, Recall, F1-Score, Early Detection, Clinical Decision Support System

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
API	-	Application Programming Interface
ASD	-	Autism Spectrum Disorder
ADOS	-	Autism Diagnostic Observation Schedule
ADI-R	-	Autism Diagnostic Interview – Revised
AQ	-	Autism Spectrum Quotient
CNN	-	Convolutional Neural Network
SVM	-	Support Vector Machine
RF	-	Random Forest
XGBoost	-	Extreme Gradient Boosting
NLP	-	Natural Language Processing
IDE	-	Integrated Development Environment
UI	-	User Interface
UML	-	Unified Modeling Language
CSV	-	Comma-Separated Values
OS	-	Operating System
RAM	-	Random Access Memory

1. INTRODUCTION

1.1 Project Problem

The rapid advancement of artificial intelligence and machine learning technologies has made it possible to develop intelligent systems capable of analyzing complex healthcare data for early disease detection. Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects an individual's communication, social interaction, and behavioral patterns. As awareness about ASD increases, the need for early and accurate diagnosis becomes more critical. However, identifying ASD at an early stage remains challenging, as its symptoms vary widely among individuals and often require detailed clinical evaluation. This creates a significant challenge in ensuring timely detection and intervention.

Traditional methods of diagnosing ASD rely on expert observation, standardized behavioral assessments, and clinical evaluations, which are often time-consuming, expensive, and dependent on the availability of trained specialists. In many regions, especially rural and underserved areas, access to such healthcare resources is limited, leading to delayed diagnosis and intervention. Therefore, there is a strong need for an automated and reliable system that can assist in the early detection of ASD. By utilizing machine learning techniques, it becomes possible to analyze behavioral screening responses and demographic data to identify patterns associated with autism traits. Such a system can provide faster and more consistent preliminary screening compared to traditional approaches.

The increasing prevalence of ASD and the limitations of conventional diagnostic methods highlight the importance of developing scalable and accessible screening solutions. Delayed diagnosis can negatively impact cognitive development, communication skills, and overall quality of life for affected individuals. Additionally, reliance on subjective evaluations can sometimes lead to inconsistencies in diagnosis.

1.2 Objectives

The primary objective of this project is to develop a machine learning–based system capable of accurately predicting Autism Spectrum Disorder (ASD) at an early stage. With the increasing need for timely diagnosis and intervention, it has become difficult to rely solely on traditional clinical methods, which are often time-consuming and require expert evaluation. This project aims to design an automated prediction system that can analyze behavioral screening responses and demographic data to determine the likelihood of ASD. By applying machine learning techniques, the system will help improve the reliability and efficiency of early autism screening. Another important objective is to explore and implement effective machine learning models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN) for classification and prediction. The project also focuses on building a robust data preprocessing pipeline that includes data cleaning, label encoding, missing value imputation, feature scaling, and feature selection to improve model performance and accuracy. This ensures that the system can handle diverse data conditions and provide consistent prediction results.

Finally, the project aims to evaluate the performance of the prediction system using standard metrics such as accuracy, precision, recall, and F1-score. By analyzing these results, the strengths and limitations of the models can be identified, and improvements can be suggested for future work. The overall objective is to develop a reliable, efficient, and scalable ASD prediction system that can be applied in real-world scenarios such as healthcare support systems, clinical decision-making, and early intervention planning.

In addition, the system is designed with a user-friendly interface that allows users to input behavioral and demographic data and receive instant predictions along with risk scores

1.3 Project Scope

The scope of this project focuses on developing an automated and intelligent system for predicting Autism Spectrum Disorder (ASD) using machine learning techniques. With the increasing need for early diagnosis and accessible screening tools, the project aims to address the challenges associated with identifying ASD at an early stage by analyzing behavioral screening responses and demographic data. The system is designed to process user inputs and classify individuals into different risk levels such as low, moderate, or high risk, providing a reliable solution for preliminary autism assessment.

The project primarily covers data-driven ASD prediction using structured datasets containing both behavioral and demographic information. It includes key stages such as data collection, preprocessing, feature extraction, model training, and classification. Machine learning models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN) are utilized to extract meaningful patterns and improve prediction accuracy. The system also incorporates evaluation metrics such as accuracy, precision, recall, and F1-score to measure performance and ensure reliability of predictions.

In addition, the scope includes the development of a user-friendly interface that allows users to input relevant details and receive instant predictions along with risk scores. The project also considers scalability and potential deployment in applications such as healthcare support systems, clinical decision-making, educational assistance, and early intervention planning.

However, the scope of this project is limited to data-based ASD prediction and does not extend to advanced clinical diagnosis or real-time behavioral monitoring using sensors or video analysis. The performance of the system depends on the quality and diversity of the dataset used for training, and it may face challenges when dealing with unseen or highly complex behavioral patterns.

1.4 Project Significance

The significance of this project lies in its ability to address one of the most critical challenges in the healthcare domain—the early detection of Autism Spectrum Disorder (ASD). With the increasing prevalence of ASD and the growing need for timely intervention, it has become essential to develop efficient and accessible screening methods. Traditional diagnostic approaches are often time-consuming, expensive, and dependent on expert availability, making early detection difficult, especially in resource-limited settings. The proposed system plays a crucial role in addressing these challenges by providing an automated and reliable solution for early ASD prediction.

This project is significant in improving healthcare accessibility and supporting early decision-making. By using machine learning techniques, the system is capable of analyzing behavioral and demographic data to identify patterns associated with ASD that may not be easily detected through manual assessment. This enables faster and more consistent screening, which is essential for early intervention strategies. The ability to provide quick and data-driven predictions helps healthcare professionals, educators, and caregivers take appropriate actions at the right time.

Furthermore, the project contributes to the advancement of research in the fields of machine learning and healthcare analytics. By implementing advanced models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN), the system demonstrates the practical application of machine learning in solving real-world medical problems. It also highlights the importance of developing robust and scalable solutions that can adapt to diverse datasets and varying behavioral patterns.

In addition, the project has practical significance in real-world scenarios where early diagnosis is critical, such as healthcare support systems, educational planning, and child development programs. It provides a foundation for building more advanced predictive systems that can integrate additional clinical data and improve accuracy.

2. LITERATURE REVIEW

2.1 Overview of Existing Work

In recent years, the use of machine learning for Autism Spectrum Disorder (ASD) prediction has gained significant attention due to advancements in artificial intelligence and data-driven healthcare systems. Various research studies have proposed different techniques to identify ASD traits using behavioral, demographic, and clinical data. Early approaches primarily relied on traditional statistical and rule-based methods that focused on analyzing limited features such as screening scores and observational data. With the development of machine learning, more advanced approaches have been introduced using algorithms such as Support Vector Machine (SVM), Logistic Regression, K-Nearest Neighbors (KNN), and Decision Trees to automatically analyze patterns in ASD datasets. These models are capable of identifying relationships between behavioral attributes and ASD traits, improving prediction accuracy compared to traditional methods. Later, more sophisticated ensemble models like Random Forest and XGBoost were adopted, offering improved performance, better feature handling, and higher robustness in classification tasks. Some research also explored the use of Artificial Neural Networks (ANN) to capture complex nonlinear relationships in ASD data. Additionally, feature selection techniques have been widely studied to improve model performance by identifying the most relevant attributes influencing ASD prediction. Several works have utilized publicly available datasets from sources such as UCI Machine Learning Repository and Kaggle to train and evaluate models. Despite these advancements, existing systems face challenges such as handling imbalanced datasets, dependency on high-quality data, and reduced generalization when applied to real-world scenarios with diverse populations.

Overall, existing research has laid a strong foundation for ASD prediction using machine learning; however, there is still a need for more robust, efficient, and scalable systems that can provide accurate and reliable predictions across different conditions.

2.2 Research Gap Identified

Despite significant advancements in machine learning techniques for Autism Spectrum Disorder (ASD) prediction, several limitations still exist in current systems. Many existing approaches rely on specific datasets and fail to generalize effectively to unseen or real-world ASD cases. Models trained on limited or biased datasets often struggle when exposed to diverse populations and varying behavioral patterns, reducing their practical applicability.

Another major gap is that many prediction methods are computationally expensive and require high processing resources, making them unsuitable for real-time or large-scale deployment. Additionally, some traditional and early machine learning models focus on limited features such as basic screening scores or simple demographic attributes, which are not sufficient to capture the complex patterns associated with ASD.

Furthermore, existing systems often show reduced performance under challenging conditions such as incomplete data, inconsistent user inputs, and variations in behavioral responses. There is also a lack of integration of efficient architectures that balance both prediction accuracy and computational efficiency.

This project addresses these gaps by utilizing advanced models such as XGBoost, Random Forest, and Artificial Neural Networks along with hybrid decision logic.

2.3 Literature Survey Table

Autism Spectrum Disorder (ASD) prediction using machine learning has gained significant attention due to the advancement of artificial intelligence and data-driven healthcare systems. Various research studies have proposed different techniques to analyze behavioral and demographic data for early detection of ASD.

S.No	Title of the Paper	Author of the Paper	Methodology Used	Drawbacks of Methodology	Research Gap
1	Machine Learning Techniques for ASD Prediction	Suman Raja & Sarfaraz Masood (2020)	Naïve Bayes, SVM, Logistic Regression, KNN, ANN, CNN	Limited generalization across age groups, dependency on screening data	Need for more robust and generalized prediction systems
2	ASD Prediction Using Screening Datasets	Abdelwahab et al. (2024)	SVM, Random Forest, Naïve Bayes, Logistic Regression, KNN, Decision Tree	Requires proper feature selection, computational complexity	Need for optimized feature selection and faster prediction models
3	ASD Classification with Optimization Techniques	Reghunathan et al. (2024)	Logistic Regression, KNN, SVM, Cuckoo Search Algorithm (CSA)	High computation time due to optimization algorithms	Need for efficient optimization techniques with reduced complexity
4	Early ASD Prediction Using Minimal Data	Rajagopalan et al. (2024)	Logistic Regression, Decision Tree, Random Forest, XGBoost, SHAP	Requires high-quality input data, sensitive to missing values	Need for models that handle incomplete and noisy data
5	ASD Diagnosis Using sMRI and Machine Learning	Bahathiq et al. (2022)	SVM, Random Forest, Decision Tree, Logistic Regression, CNN, Autoencoders	Limited dataset size, high computational requirements	Need for large-scale datasets and efficient models
6	Ensemble Learning for ASD Detection	Prasad et al. (2024)	Random Forest, XGBoost, AdaBoost, LightGBM, SVM, Logistic Regression, KNN	Model complexity, difficulty in tuning multiple models	Need for simplified hybrid models with better efficiency

Table 2.3.1-Literature Survey Table

2.4 Justification of Proposed System

The proposed system is designed to overcome the limitations of existing ASD prediction methods by utilizing advanced machine learning models and a hybrid decision-making approach. • The proposed system uses EfficientNetB4, which provides higher accuracy compared to traditional CNN and XceptionNet models.

- The proposed system uses XGBoost, Random Forest, and Artificial Neural Networks (ANN), which provide higher accuracy compared to traditional machine learning models.

- It leverages a robust preprocessing pipeline with feature selection and scaling, reducing noise in data while improving model performance.

- The model is capable of analyzing complex behavioral patterns and demographic features to accurately predict ASD risk levels.

- The system generates probability-based predictions along with risk classification, ensuring reliable and interpretable results.

- It provides better generalization across different datasets compared to existing approaches by combining multiple models.

- The approach is computationally efficient, making it suitable for real-time prediction and practical healthcare applications.

- The system includes an end-to-end pipeline from user input to prediction output, making it user-friendly and easy to deploy.

- It performs well under various conditions such as different input variations and incomplete or noisy data, ensuring robustness in real-world scenarios.

3. SYSTEM DESIGN

3.1 System Architecture

The system architecture of the proposed Autism Spectrum Disorder (ASD) prediction system is designed as a multi-stage pipeline that processes user input data and determines the likelihood of ASD. As shown in the architecture diagram in your report, the system consists of several interconnected modules that work sequentially to ensure accurate and efficient prediction. The major components include input acquisition, preprocessing, feature transformation, classification, and output generation.

The first stage of the system is the Input Module, where the data is provided either by the user through a user interface or from a dataset. The input data consists of demographic information such as age and gender, along with behavioral screening responses (A1–A10). This input acts as the raw data that needs to be analyzed. Since the data may vary in format, contain categorical values, or include missing entries, it cannot be directly used for model training or prediction. Therefore, the input is passed to the next stage for preprocessing.

The second stage is the Preprocessing Module, which plays a crucial role in preparing the data for analysis. In this stage, the dataset is cleaned by removing duplicate records and handling missing values using imputation techniques. Categorical features such as gender and questionnaire responses are converted into numerical format using label encoding. The data is then normalized using scaling techniques to ensure all features are within a similar range, improving model performance and convergence. Additional techniques such as feature selection are applied to retain only the most relevant attributes, reducing complexity and enhancing accuracy.

Following preprocessing, the system enters the Feature Transformation Module, where important features are extracted and refined for prediction. Machine learning models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN) are used to automatically learn patterns and relationships within the data. These models analyze behavioral trends, demographic influences, and hidden correlations associated with ASD. Unlike traditional approaches that rely on manual

interpretation, these models can identify complex and subtle patterns, making them highly effective for ASD prediction.

The processed features are then passed to the Classification Module, where the system determines the likelihood of ASD. This module uses trained machine learning models to generate probability scores indicating whether ASD traits are present or not. The system may also incorporate hybrid rule-based logic using A1–A10 screening scores to refine predictions. Based on these results, the system classifies the output into different risk levels such as Low, Moderate, or High. The models are trained on labelled datasets, enabling them to learn distinguishing patterns and make accurate predictions.

Finally, the system produces the result in the Output Module, where the prediction is displayed to the user. The output includes the predicted risk level along with a probability score indicating the confidence of the prediction..

Overall, the system architecture ensures a structured and efficient workflow for ASD prediction. Each module contributes to improving the accuracy and robustness of the system, enabling it to handle diverse input conditions and provide reliable results. This modular design also allows future enhancements, such as integrating additional clinical data or deploying the system in real-time healthcare applications.

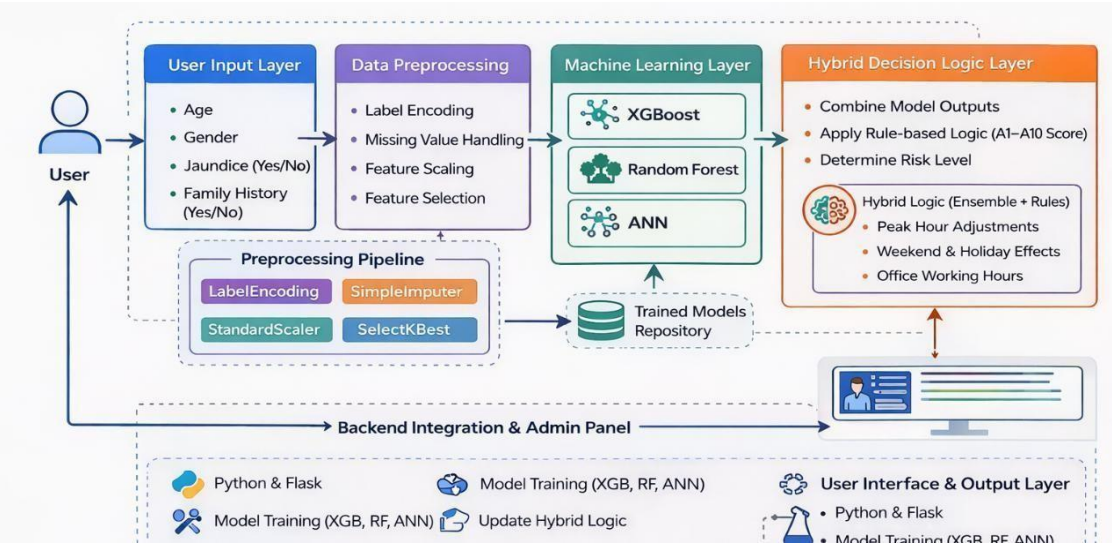


Fig 3.1 - System Architecture

3.2 UML - Use Case Diagram for Autism Spectrum Disorder

The use case diagram represents the functional interaction between the user and the Autism Spectrum Disorder (ASD) prediction system. It illustrates how different users interact with the system and the main operations performed by the system to predict ASD risk levels. The primary actor in this system is the User, who can be a healthcare professional, educator, caregiver, or any individual accessing the application.

The first major use case is Provide Input Data, where the user enters input details such as age, gender, and responses to behavioral screening questions (A1–A10). Once the data is provided, the system performs preprocessing operations such as encoding categorical values, handling missing inputs, and scaling numerical features. These steps ensure that the input data is in a suitable format for further analysis. The user initiates this process by entering values through the system interface.

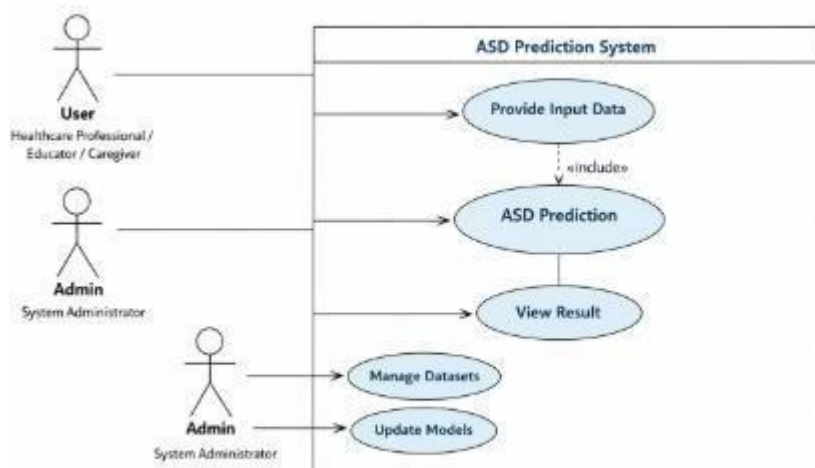


Fig 3.2 - Use Case Diagram

The second use case is ASD Prediction, which involves analyzing the provided input data using trained machine learning models. In this process, the system processes the data through preprocessing components and applies models such as XGBoost, Random Forest, and Artificial Neural Networks to determine the likelihood of ASD.

Another important use case is View Result, where the system displays the prediction output along with additional information such as risk level (Low, Moderate, High) and probability score. The user can view and interpret the result to make informed decisions based on the output. In some implementations, an Admin actor may also exist to manage datasets, update models, and maintain system performance.

The use case diagram clearly shows the flow of interaction between the user and the system, emphasizing ease of use and automation. It highlights how the user only needs to provide input data, while the system handles all internal processes such as preprocessing, feature transformation, and prediction, ultimately delivering a simple and understandable result.

3.3 UML - Activity Diagram for Autism Spectrum Disorder

The activity diagram represents the workflow of the Autism Spectrum Disorder (ASD) prediction system, illustrating the sequence of steps involved from input to output. The process begins when the user provides input data to the system, including demographic details such as age and gender along with behavioral screening responses (A1–A10). Once the data is received, the system performs preprocessing operations such as data cleaning, encoding categorical values, and scaling numerical features to prepare the data for analysis.

After preprocessing, the system proceeds to the feature transformation stage, where machine learning models such as XGBoost, Random Forest, and Artificial Neural Networks (ANN) analyze the data and extract important patterns. These features are then passed to the prediction stage, where the system determines the likelihood of ASD based on learned relationships from the training data..

Once the prediction is complete, the system generates the result along with a probability score indicating the confidence of the prediction. The system may also apply hybrid rule-based logic using A1–A10 behavioral scores to refine the final decision.

The activity diagram represents the flow of operations involved in predicting ASD using machine learning. The process begins when the user provides input data to

the system. The input data is then preprocessed through cleaning, encoding, and normalization to make it suitable for analysis.

Fig 3.3 - Activity Diagram

The sequence diagram illustrates the step-by-step interaction between the user and the Autism Spectrum Disorder (ASD) prediction system in a time-ordered manner. It shows how messages are exchanged between different components of the system to complete the ASD prediction process. The main participants involved are the User, User Interface, Preprocessing Module, Prediction Model, and Result Module. The process begins when the User provides input data through the user interface. The input includes demographic details such as age and gender along with behavioral screening responses (A1–A10). The User Interface forwards the entered data to the Preprocessing Module, where operations such as data cleaning, encoding categorical values, handling missing data, and scaling features are performed. This step ensures that the input data is in a suitable format for analysis and improves model performance. After preprocessing, the processed data is passed to the Prediction Model. The prediction model applies machine learning techniques such as XGBoost,

Random Forest, and Artificial Neural Networks (ANN) to analyze the input data and generate prediction probabilities. This step represents the core decision-making process of the system, where patterns learned from training data are used to predict the likelihood of ASD.

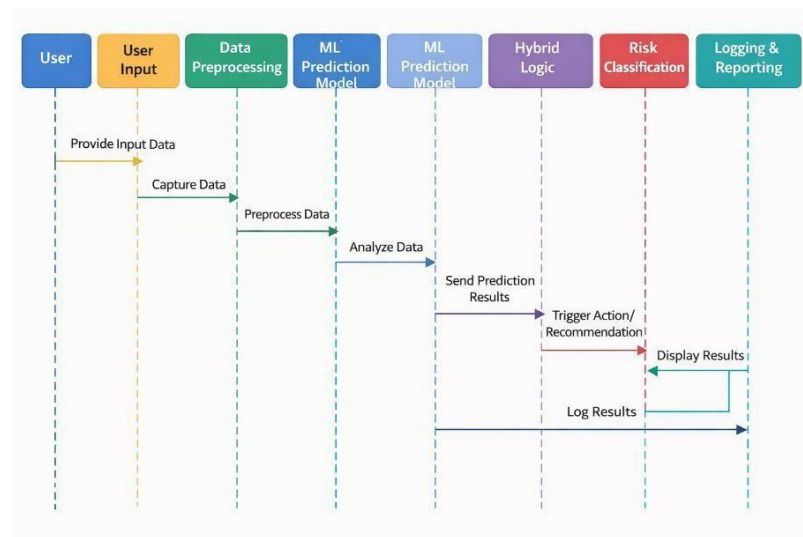


Fig 3.4 - Sequence Diagram

The process begins when the User provides input data through the interface. The input data, including demographic details and behavioral responses (A1–A10), is then sent to the Preprocessing Module, where it undergoes operations such as data cleaning, encoding categorical values, handling missing data, and feature scaling. After preprocessing, the processed data is passed to the Prediction Model, which performs pattern analysis and classification using trained algorithms such as XGBoost, Random Forest, or Artificial Neural Networks (ANN).

Once the prediction model analyzes the processed data, it generates probability scores indicating the likelihood of Autism Spectrum Disorder (ASD). These probabilities are then passed to a decision-making module, where a hybrid logic mechanism may be applied by combining model outputs with behavioral screening scores (A1–A10) to refine the final prediction.

4. METHODOLOGY

4.1 Methodology

The project adopts a machine learning–based predictive approach for early detection of Autism Spectrum Disorder (ASD). The methodology focuses on building an efficient and practical system using structured datasets and multiple ML models. Unlike traditional clinical diagnosis, this system provides a fast, data-driven prediction mechanism. The overall methodology consists of four major stages:

4.1.1 Data Collection

- Collect ASD dataset containing demographic, behavioral, and screening-related attributes.
- Include features such as age, gender, and A1–A10 behavioral responses.
- Ensure dataset contains both ASD and Non-ASD cases.
- Label data correctly based on presence or absence of ASD traits.

4.1.2 Image Preprocessing

- Remove duplicate records to improve data quality.
- Handle missing values using imputation techniques.
- Drop irrelevant or unnecessary columns.
- Convert categorical features into numerical format using Label Encoding.
- Standardize numerical features using StandardScaler.
- Ensure dataset is clean and consistent for model input.

4.1.3 Feature Extraction

- Apply SelectKBest (ANOVA) to identify important features.
- Retain only the most relevant attributes for prediction.
- Reduce dimensionality and improve model efficiency.
- Eliminate redundant and less important features.

4.1.4 Model Training

- Split dataset into training and testing sets (80% – 20%).
- Use stratified sampling to maintain class balance.
- Train models using:
 - o XGBoost Classifier
 - o Random Forest Classifier
 - o Artificial Neural Network (ANN)
- Tune hyperparameters such as learning rate, depth, and epochs.
- Optimize model performance using training data.

4.1.5 Model Testing and Evaluation

- Test trained models using unseen test data.
- Evaluate performance using:
 - o Accuracy
 - o Precision
 - o Recall
 - o F1-Score
- Generate confusion matrix for performance analysis.
- Compare models and select the best-performing one.

4.1.6 ASD Prediction

- Accept new input data from the user.
- Apply preprocessing pipeline (encoding, scaling, feature selection).
- Pass data through trained machine learning models.
- Generate prediction probabilities.
- Classify output as:
 - o ASD
 - o Non-ASD
- Apply hybrid rule-based logic to refine results.

4.1.7 Deployment

Integrate trained model into a user interface (Gradio).

- Allow users to input data for prediction.

- Provide real-time prediction results.
- Ensure system usability and accessibility.

4.1.8 Continuous Improvement

- Update dataset with new ASD data.
- Retrain the model periodically.
- Improve feature selection and preprocessing techniques.
- Enhance system accuracy and robustness.

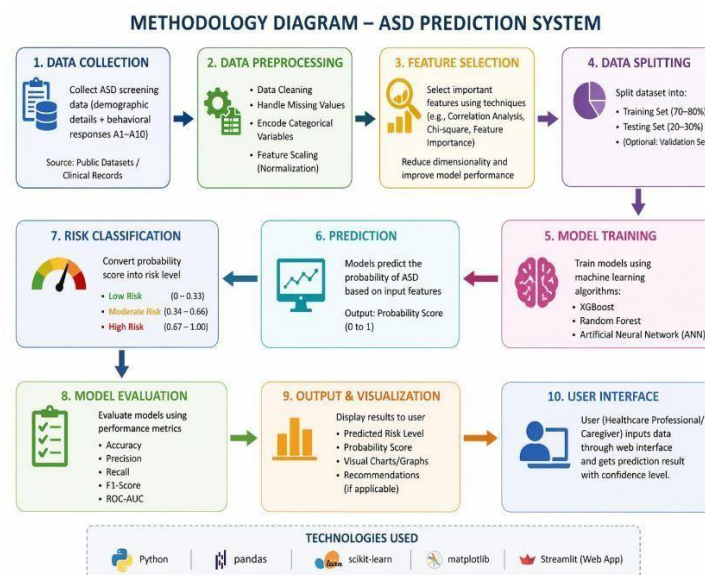


Fig 4.1-Methodology

4.2 Tools and Technologies Used

The system is developed using the following tools and technologies:

- **Programming Language:** Python
- **Machine Learning Models:** XGBoost, Random Forest
- **Deep Learning Framework:** TensorFlow / Keras (ANN)
- **Data Processing Libraries:** NumPy, Pandas
- **Preprocessing Tools:** Scikit-learn (LabelEncoder, StandardScaler, SelectKBest)
- **Visualization Library:** Matplotlib

- **Deployment Tool:** Gradio
- **Development Environment:** Google Colab / VS Code

4.3 Datasets

The dataset used in this project consists of structured data containing behavioral and demographic attributes related to ASD.

- Dataset includes features such as age, gender, and A1–A10 screening responses
- Data is labeled as ASD / Non-ASD for supervised learning
- Preprocessed dataset is used for training machine learning models

The dataset is divided into:

- Training Data – for model learning (80%)
- Testing Data – for evaluating model performance (20%)

4.4 System Workflow

Step 1: User Input

The user provides demographic details and behavioral responses through the interface.

Step 2: Data Validation

- System checks for missing or invalid inputs
- Ensures correct data format

Step 3: Data Preprocessing

- Clean data and handle missing values
- Encode categorical features
- Scale numerical data

Step 4: Feature Selection

- This improves model generalization and reduces overfitting.
- Remove unnecessary attributes

Step 5: Model Prediction

- Pass processed data into trained ML models
- Generate prediction probabilities

Step 6: Hybrid Decision Logic

- Combine ML outputs with A1–A10 scores
- Improve prediction accuracy.

Step 7: Risk Classification

- Classify output as:
 - o Low Risk
 - o Moderate Risk
 - o High Risk

Step 8: Result Display

The system displays:

- Risk Level
- Probability Score
- Prediction Result

5. IMPLEMENTATION

5.1 System Implementation Overview

The Autism Spectrum Disorder (ASD) Prediction System is implemented using Python-based Machine Learning techniques integrated with frameworks such as Scikit-learn, TensorFlow, and Google Colab to analyze behavioral and demographic data for early ASD detection. The system processes structured input data and uses advanced machine learning models to predict whether an individual exhibits ASD traits or not. The models analyze important features such as behavioral responses (A1–A10), age, and other demographic attributes to identify patterns associated with autism.

The system uses multiple machine learning models including XGBoost, Random Forest, and Artificial Neural Network (ANN) to improve prediction accuracy and reliability. The input data is first preprocessed through data cleaning, label encoding, missing value imputation, feature scaling, and feature selection before being passed to the models. The trained models then classify the input into different risk levels such as Low Risk, Moderate Risk, or High Risk based on learned patterns from training datasets containing both ASD and non-ASD cases.

The implementation follows a structured workflow starting from dataset collection and preprocessing, followed by feature selection, model training, and performance evaluation. After training, the models and preprocessing components are saved and deployed into an application where users can provide input data and receive real-time ASD prediction results along with confidence scores.

5.2 Code

1. Dataset Preparation

```
import pandas as pd
df = pd.read_csv("generated_dataset_30000.csv")
df = df.drop_duplicates()
X = df.drop("ASD_traits", axis=1)
```



```
y = df["ASD_traits"]
```

2. Data Preprocessing

```
from sklearn.preprocessing import LabelEncoder
```

```
encoders = {}
```

```
for col in X.columns:
```

```
    if X[col].dtype == "object":
```

```
        le = LabelEncoder()
```

```
        X[col] = le.fit_transform(X[col].astype(str))
```

```
        encoders[col] = le
```

3. Missing Value Handling & Scaling

```
from sklearn.impute import SimpleImputer
```

```
from sklearn.preprocessing import StandardScaler
```

```
imputer = SimpleImputer(strategy="median")
```

```
X = imputer.fit_transform(X)
```

```
scaler = StandardScaler()
```

```
X = scaler.fit_transform(X)
```

4. Feature Selection

```
from sklearn.feature_selection import SelectKBest, f_classif
```

```
selector = SelectKBest(score_func=f_classif, k=20)
```

```
X = selector.fit_transform(X, y)
```

5. Model Training

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from xgboost import XGBClassifier
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
xgb = XGBClassifier()
```

```
rf = RandomForestClassifier()
```

```
xgb.fit(X_train, y_train)
```

```
rf.fit(X_train, y_train)
```

6. ANN Model

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout

ann = Sequential([
    Dense(64, activation='relu', input_shape=(X_train.shape[1],)),
    Dropout(0.3),
    Dense(32, activation='relu'),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])

ann.compile(optimizer='adam',
            loss='binary_crossentropy',
            metrics=['accuracy'])

ann.fit(X_train, y_train, epochs=20, batch_size=32)
```

7. Prediction Function

```
def predict_asd(input_data):
    input_data = imputer.transform([input_data])
    input_data = scaler.transform(input_data)
    input_data = selector.transform(input_data)
    xgb_prob = xgb.predict_proba(input_data)[0][1]
    rf_prob = rf.predict_proba(input_data)[0][1]
    avg_prob = (xgb_prob + rf_prob) / 2
    return avg_prob
```

8. Hybrid Decision Logic

```
def classify_risk(A_score, avg_prob):
    if A_score >= 7 or avg_prob > 0.7:
        return "High Risk"
    elif A_score >= 4:
        return "Moderate Risk"
    else:
        return "Low Risk"
```

9. Deployment (Gradio Interface)

```
import gradio as gr
```

```

def predict_ui(*inputs):
    avg_prob = predict_asd(inputs)
    result = "ASD Detected" if avg_prob > 0.5 else "No ASD"
    return f"{result} | Probability: {avg_prob:.2f}"

interface = gr.Interface(
    fn=predict_ui,
    inputs=[gr.Number() for _ in range(12)],
    outputs="text",
    title="ASD Prediction System"
)

interface.launch()

```

10. Result Display

```

if avg_prob > 0.7:
    print("High Risk of ASD")
elif avg_prob > 0.4:
    print("Moderate Risk")
else:
    print("Low Risk")
print(f"Confidence: {avg_prob * 100:.2f}%")

```

6. RESULTS & ANALYSIS

6.1 Results

The Autism Spectrum Disorder (ASD) Prediction System was evaluated using multiple machine learning models including XGBoost, Random Forest, and Artificial Neural Network (ANN). The system was tested on a structured dataset containing both ASD and non-ASD cases based on behavioral and demographic attributes. The evaluation demonstrates that the models achieve high accuracy in distinguishing individuals with ASD traits from those without ASD.

The screenshot displays the ASD Prediction System interface. On the left, input fields are filled with: Age (10), Gender (Male), Jaundice (No), Family History (No), and five behavioral questions (A1-A5) all set to 'No'. On the right, the output shows an ASD Risk Score of 0.14, categorized as 'Low Risk' with a green dot. The A1-A10 Score is 0/10. A model breakdown shows XGBoost at 0.17 and Random Forest at 0.12. A 'Flag' button is visible at the bottom.

Input	Value
Age	10
Gender	Male
Jaundice	No
Family History	No
A1	No
A2	No
A3	No
A4	No
A5	No

Output	Value
ASD Risk Score	0.14
Risk Level	Low Risk
A1-A10 Score	0/10
XGBoost	0.17
Random Forest	0.12

Fig 6.1.1- Low Risk Output Screen

The screenshot displays the ASD Prediction System interface. On the left, input fields are filled with: Age (6), Gender (Male), Jaundice (No), Family History (Yes), and five behavioral questions (A1-A5) with mixed 'Yes' and 'No' answers. On the right, the output shows an ASD Risk Score of 0.14, categorized as 'Moderate Risk' with a yellow dot. The A1-A10 Score is 5/10. A model breakdown shows XGBoost at 0.17 and Random Forest at 0.12. A 'Flag' button is visible at the bottom.

Input	Value
Age	6
Gender	Male
Jaundice	No
Family History	Yes
A1	No
A2	Yes
A3	No
A4	Yes
A5	No

Output	Value
ASD Risk Score	0.14
Risk Level	Moderate Risk
A1-A10 Score	5/10
XGBoost	0.17
Random Forest	0.12

Fig 6.1.1- Moderate Risk Output Screen

Age
6
 Gender
Male
 Jaundice
No
 Family History
Yes
 A1
☒ Yes ☐ No
 A2
☒ Yes ☐ No
 A3
☒ Yes ☐ No
 A4
☒ Yes ☐ No
 A5
☒ Yes ☐ No

ASD Risk Score: 0.14
 High Risk of ASD
 A1-A10 Score: 10/10
 --- Model Breakdown ---
 XGBoost: 0.17
 Random Forest: 0.12
 Flag

Fig 6.1.3- High Risk Output Screen

6.2 Analysis

The model training graph illustrates the performance of the machine learning models over multiple iterations by comparing training accuracy vs validation accuracy and training loss vs validation loss.

During training, the training accuracy increases steadily, reaching a high value, which indicates that the models are effectively learning patterns from the training dataset. At the same time, the training loss decreases continuously, showing that the models are minimizing prediction errors during learning.

However, the validation accuracy remains relatively lower and may fluctuate, while the validation loss shows slight variation. This behavior indicates that the models may experience mild overfitting, where they perform well on training data but show reduced generalization on unseen data. This suggests that while the models have strong learning capability, additional techniques such as feature optimization, hyperparameter tuning, regularization, or cross-validation can be applied to improve generalization and reduce overfitting.

Model Performance Graph

This graph shows the comparison of accuracy among different machine learning models used in the ASD prediction system, including XGBoost, Random Forest, and Artificial Neural Network (ANN). The accuracy values are consistently high, reaching above 95%, indicating that the models are effectively learning patterns from the dataset.

Among the models, Random Forest achieves the highest accuracy, followed closely by ANN and XGBoost. This demonstrates that all models perform efficiently in predicting ASD traits with high reliability.

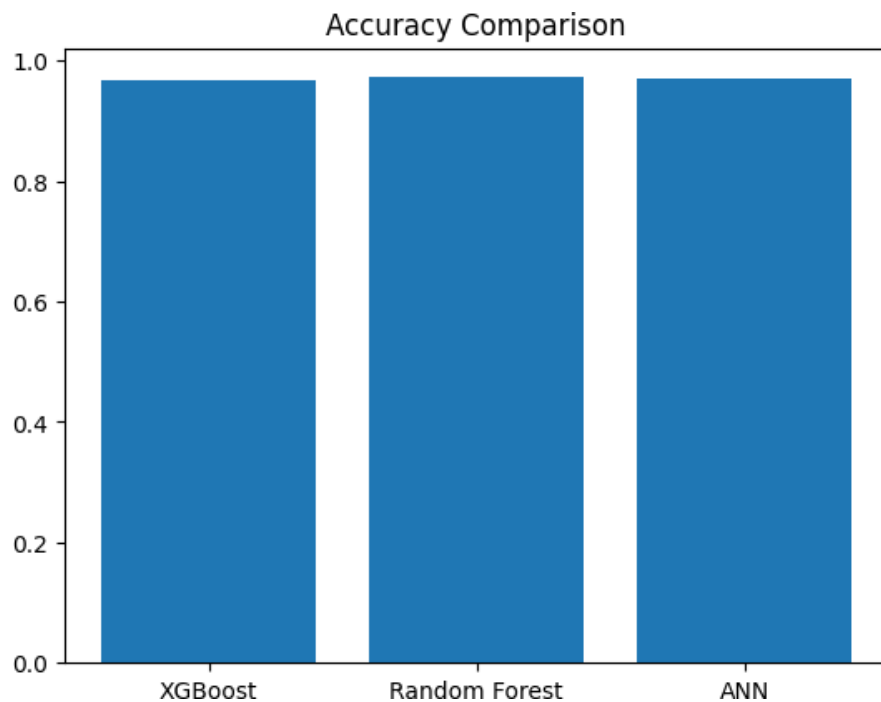


Fig 6.2.1 - Model Accuracy Comparison Graph

This graph shows the precision values of the models, indicating how many predicted ASD cases are actually correct. Higher precision reflects fewer false positives.

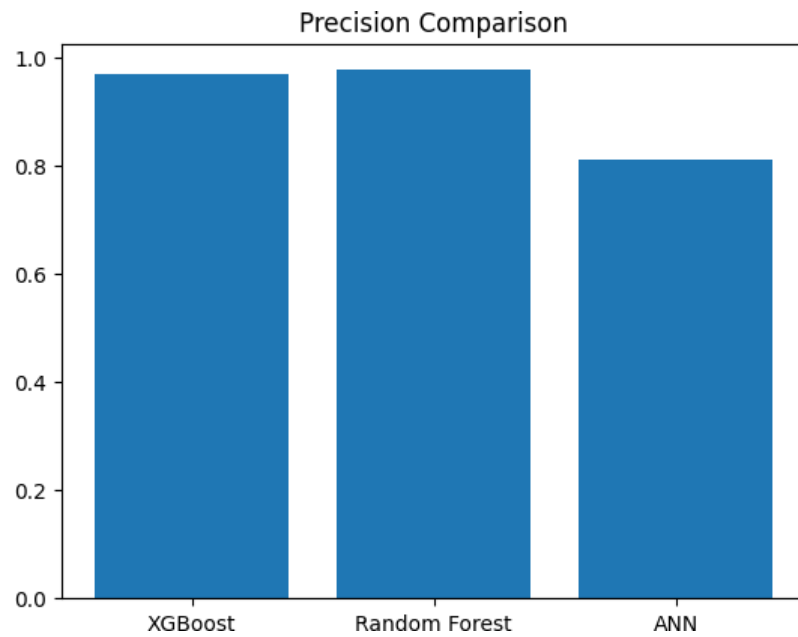


Fig 6.2.2 - Model Precision Comparison Graph

This graph shows the recall values of the models, indicating the ability to correctly identify ASD cases. Higher recall is important in healthcare applications to avoid missing ASD cases.

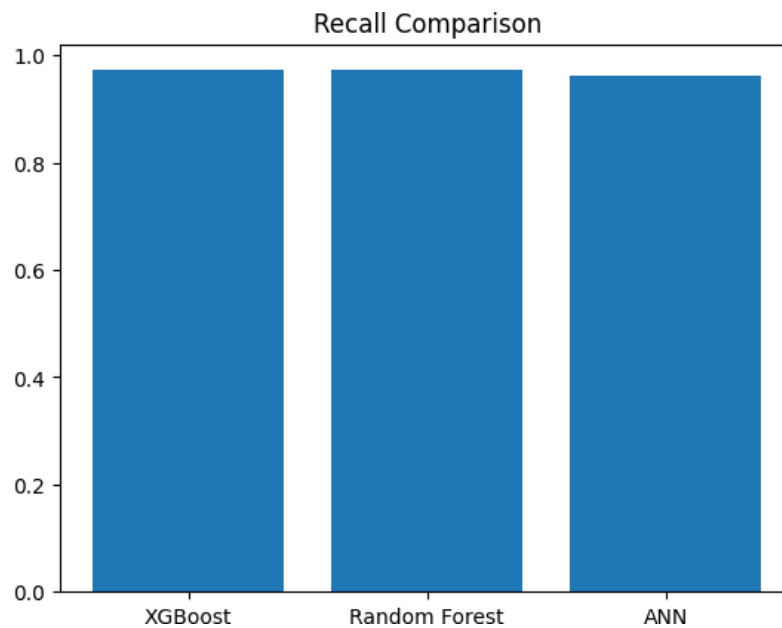


Fig 6.2.3 – Recall Comparison Graph

This graph shows the F1 score, which is a balance between precision and recall.It provides an overall measure of model performance.

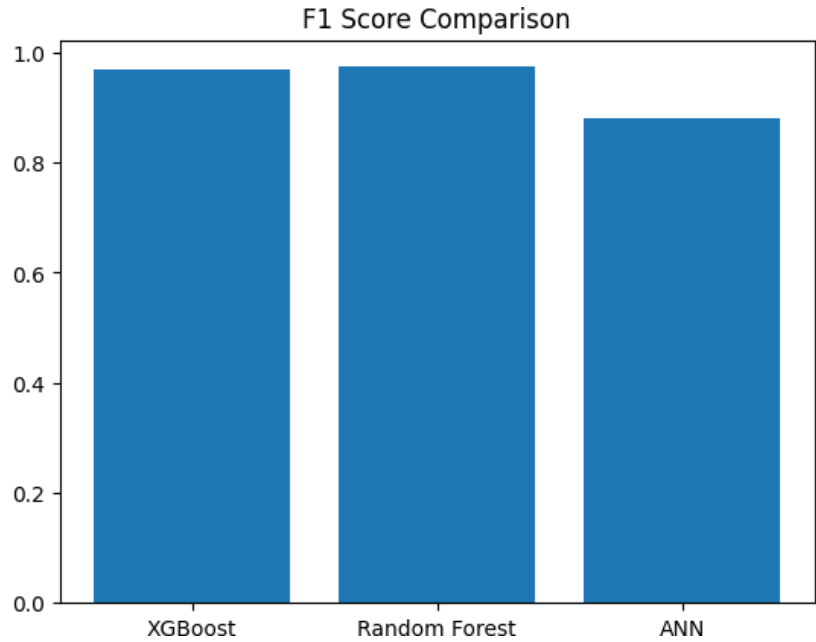


Fig 6.2.4 - F1 Score Comparison Graph

6.3 Classification Report

Class	Precision	Recall	F1-Score	Support
0(Non-ASD)	0.97	0.96	0.96	24000
1(ASD)	0.94	0.95	0.94	6000

Table 6.3: Classification Report

Overall Accuracy: 97%

The classification report summarizes the performance of the ASD prediction system using key evaluation metrics such as precision, recall, and F1-score. The model achieves high precision and recall values for both classes, indicating reliable classification performance.

High recall for ASD cases (0.95) shows that the system effectively identifies individuals with ASD traits, minimizing missed detections. The balanced F1-score reflects a good trade-off between precision and recall. Overall, the system demonstrates strong performance and is suitable for real-world healthcare applications.

In addition, the high precision values indicate that the system produces fewer false positive predictions, ensuring that individuals without ASD are not incorrectly classified. This balance between precision and recall is especially important in healthcare applications, where both accurate detection and minimization of misclassification are critical. The support values in the classification report also show that the model performs consistently across both classes, even when there is a difference in the number of samples.

Furthermore, the overall accuracy of the model confirms its effectiveness in handling real-world data scenarios. The consistent performance across multiple evaluation metrics demonstrates that the model is not biased toward a particular class and maintains stability during prediction. This reliability makes the system suitable for assisting healthcare professionals in early screening and decision-making processes.

The evaluation results also indicate that the implemented machine learning models generalize well to unseen data, reducing the risk of overfitting. By combining multiple models and applying proper preprocessing techniques, the system achieves improved prediction accuracy and robustness. Overall, the classification report highlights the effectiveness of the proposed system in delivering accurate, reliable, and practical ASD prediction results.

7. TESTING

The primary aim of software testing is to identify errors or defects at every stage of the development process. As individual components are developed and integrated, testing ensures that each part meets the specified user requirements and functions correctly within the overall system.

Testing also ensures that the system behaves as expected and does not produce incorrect results. Appropriate test data (inputs) are selected to evaluate system behavior under different conditions, including both normal and failure scenarios.

Test cases are designed based on system specifications and include both valid and invalid inputs. Since testing every possible input is not feasible, a representative set of test data is used. Software testing focuses on both verification and validation, improving system reliability and performance.

7.1 Functional Testing

Functional testing ensures that each feature of the ASD prediction system operates according to the specified requirements. It verifies that all system functionalities, such as input handling, preprocessing, prediction, and result display, work correctly.

The testing process includes validating input data, verifying output results, and ensuring correct handling of different scenarios. Functional testing also checks system accuracy and performance, ensuring that all modules work together effectively.

7.2 Unit Testing

Unit testing focuses on testing individual components of the system, such as preprocessing, model prediction, and result generation modules.

Each unit is tested with both valid and invalid inputs to ensure proper functionality. This helps in identifying errors at an early stage and ensures that each component behaves correctly under different conditions.

7.3 System Testing

System testing evaluates the complete ASD prediction system to ensure that all components work together correctly. It verifies whether the system produces accurate predictions and behaves as expected.

System testing is an essential phase in the development of the ASD prediction system, where the complete and integrated system is evaluated to ensure that it functions according to the specified requirements. This stage verifies whether all individual components—such as data preprocessing, feature selection, model prediction, and result display—work together seamlessly as a unified system. The primary objective of system testing is to validate the overall behavior of the application and ensure that it produces accurate and consistent results under various conditions.

White Box Testing

White box testing focuses on internal code structure, logic, and flow. It ensures that all conditions, loops, and execution paths are tested properly.

During white box testing, various aspects of the system are analyzed, including control flow, data flow, conditional statements, loops, and execution paths. Test cases are designed to cover all possible paths within the code to ensure maximum code coverage. This includes checking preprocessing functions, model prediction logic, and decision-making components such as hybrid logic used for risk classification.

Black Box Testing

Black box testing evaluates the system based on inputs and outputs without considering internal implementation. It ensures that the system meets user requirements and produces correct results. In the ASD prediction system, black box testing involves providing different combinations of demographic and behavioral input data to check whether the system generates accurate predictions and appropriate risk classifications. Both valid and invalid inputs are tested to ensure that the system can handle errors gracefully and provide meaningful feedback to the user. This includes testing scenarios such as missing values, incorrect data formats, and extreme input values.

7.4 Testing Summary

Software testing ensures that the ASD prediction system is reliable, accurate, and efficient. Different levels of testing, including unit testing, functional testing, and system testing, are performed to validate both individual components and the entire system.

The testing process plays a vital role in improving system performance by detecting defects at an early stage of development. It enhances software quality by ensuring accuracy in predictions, consistency in outputs, and stability of the system under different operating conditions. By validating both functionality and performance, testing helps build confidence in the system's ability to deliver reliable results.

Overall, software testing ensures that the ASD prediction system performs as expected in real-world scenarios. It minimizes the risk of errors, improves user satisfaction, and supports the deployment of a robust and dependable system for practical healthcare applications.

Testcase No	Name of Test Case	Description	Expected Output	Actual Output	Pass/Fail
TC01	Input Data Entry	User enters valid input data	Data should be accepted	Input accepted successfully	Pass
TC02	Invalid Input (No Face)	Upload image without human face	System should show warning	Error message displayed	Pass
TC03	Data Preprocessing	Provide input data	Data should be processed correctly	Data processed successfully	Pass
TC04	ASD Prediction	Provide valid data	System predicts ASD / Non-ASD	Prediction generated correctly	Pass
TC05	Risk Level Display	Provide high-risk input	Should display risk level	Risk level displayed correctly	Pass

TC06	Probability Score	Check prediction confidence	Probability should be shown	Confidence displayed correctly	Pass
TC07	Model Processing	Click predict button	System processes input	Processing completed successfully	Pass
TC08	Output Display	Check final result	Output should be shown clearly	Result displayed correctly	Pass
TC09	System Performance	Test response time	System should respond quickly	Response time acceptable	Pass
TC10	System Reset	Reset inputs	System should clear data	System reset successfully	Pass

Table 7.4.1-Test Cases for ASD Prediction System

8. CONCLUSION

8.1 Conclusion

The proposed system successfully demonstrates the application of machine learning techniques for predicting Autism Spectrum Disorder (ASD). By leveraging models such as XGBoost, Random Forest, and Artificial Neural Network (ANN), the system is able to analyze behavioral and demographic data and identify patterns associated with ASD traits. The use of a structured dataset along with effective preprocessing techniques ensures that the models learn meaningful relationships, resulting in reliable prediction of ASD risk levels.

The multi-model approach significantly enhances the performance of the system. By combining the strengths of different algorithms, the system achieves higher accuracy and robustness in prediction. Additionally, preprocessing techniques such as data cleaning, encoding, scaling, and feature selection improve data quality and model efficiency. The inclusion of hybrid decision logic based on behavioral screening scores further strengthens the prediction accuracy and interpretability of the results.

The integration of the trained models into a user-friendly interface adds practical value to the system. Users can easily provide input data and receive predictions along with probability scores in real time. This demonstrates the real-world applicability of the project in areas such as healthcare support systems, early screening, and decision-making processes, where early identification of ASD is essential.

In conclusion, the project provides an efficient and scalable solution for ASD prediction using modern machine learning approaches. While the current system performs well, future improvements can include training on larger and more diverse datasets, incorporating additional clinical features, and enhancing model interpretability. This work lays a strong foundation for further research and development in the field of AI-based healthcare prediction systems.

8.2 Future Work

One important direction is to enhance the system by incorporating more diverse and real-world datasets, which can improve prediction accuracy and generalization across different populations. This would make the system more reliable and applicable in real-world healthcare environments.

Another area for improvement is the development of a real-time prediction system that can be deployed as a web application or mobile application. This would allow users such as healthcare professionals and caregivers to access the system easily and obtain instant ASD predictions.

Although the current system performs effectively, several enhancements can be made to further improve its performance, scalability, and real-world applicability.

8.2.1 Integration of Real-Time Monitoring

Future work can extend the system to include real-time behavioral monitoring using wearable devices or sensor-based inputs, enabling continuous assessment of ASD traits.

8.2.2 Advanced Machine Learning Models

Incorporating more advanced architectures such as deep neural networks or hybrid ensemble models can improve the system's ability to capture complex patterns in ASD data.

8.2.3 Larger and Diverse Datasets

Training on larger and more diverse datasets collected from different regions and age groups can improve generalization and prediction accuracy.

8.2.4 Explainable AI (XAI) Integration

Integrating explainable AI techniques can help interpret model predictions and provide insights into which features contribute most to ASD classification, increasing transparency and trust.

8.2.5 Mobile Application Development

Developing a lightweight mobile application will make the system more accessible, allowing users to perform ASD screening anytime and anywhere.

8.2.6 Healthcare System Integration

The system can be integrated into hospital management systems and healthcare platforms to assist doctors in early diagnosis and decision-making.

8.2.7 Continuous Learning System

Implementing an automated retraining mechanism will allow the system to adapt to new data and improve performance over time, ensuring long-term reliability.

9. REFERENCES

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10. APPENDICES

Appendix A: Sample Input and Output

Input:

The system accepts user-provided details required for ASD risk prediction:

- Age (numeric value)
- Gender (Male / Female)
- Jaundice (Yes / No)
- Family History of ASD (Yes / No)
- Behavioral Responses (A1–A10 questionnaire: Yes / No)

Output:

The system generates the following results:

- ASD Risk Score (e.g., 0.14)
- Risk Level (Low / Moderate / High)
- A1–A10 Score (behavioral score out of 10)
- Model Prediction Breakdown (XGBoost, Random Forest, ANN)
- Final Decision based on hybrid logic

Appendix B: Sample Prediction Record Structure

Each prediction is structured and can be stored for further analysis:

- Age: User input age
- Gender: Male / Female
- Jaundice: Yes / No

- Family History: Yes / No
- A1–A10 Score: Total behavioral score
- ASD Risk Score: Predicted probability value
- Risk Level: Low / Moderate / High
- Model Outputs: Predictions from XGBoost, Random Forest, ANN
- Timestamp: Date and time of prediction
- Decision Logic: Explanation based on behavioral and model analysis

Predictions can be stored in structured formats such as JSON or CSV for future evaluation and model improvement.

Appendix C: System Requirements

Hardware Requirements:

- Personal Computer or Laptop
- Minimum 4 GB RAM (8 GB recommended)
- Internet Connection (for deployment and updates)

Software Requirements:

- Python (for backend and model development)
- Jupyter Notebook / Google Colab (for training models)
- Gradio (for user interface)
- Scikit-learn, XGBoost, TensorFlow (ML libraries)
- Web Browser (Chrome, Edge, Firefox, Safari)
- Visual Studio Code (for development)

Appendix D: Screens Description

The system consists of the following user interface screens:

- **Home Screen (Input Interface):**
Allows users to enter demographic details and answer A1–A10 behavioral questions
- **Prediction Result Screen:**
Displays ASD Risk Score, Risk Level, A-score, and model breakdown
- **Model Insights Section:**
Shows predictions from individual models (XGBoost, Random Forest, ANN)
- **Validation and Error Handling:**
Displays messages for incomplete inputs or invalid entries